

Ayush JAIN

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EDUCATION

Carnegie Mellon University

PhD in Robotics

Thesis Advisor: Prof. Katerina Fragkiadaki

Thesis Committee: Prof. Deva Ramanan, Prof. Shubham Tulsiani & Prof. Leonidas Guibas

Partially Supported by:

CMU Robotics Vision Fellowship (24-25 AY)

Meta AI Mentorship Fellowship (25-26 AY)

Pittsburgh, PA

Aug. 2023 – Apr. 2027 (expected)

Carnegie Mellon University

Masters in Robotics (4.0 / 4.0)

Thesis Advisor: Prof. Katerina Fragkiadaki

Thesis Committee: Prof. Tom Mitchell, Prof. Shubham Tulsiani & Nikolaos Gkanatsios

Pittsburgh, PA

Aug. 2021 – Aug. 2023

Birla Institute of Technology and Science (BITS Pilani)

Bachelor in Computer Science (9.33 / 10.0)

Thesis: Active Embodied Vision - Towards Self-Supervised Never Ending Learners

Thesis Advisor: Prof. Katerina Fragkiadaki & Prof. Pratik Narang

Pilani, India

Aug. 2017 – May 2021

POSITIONS

Aug. 2021 Present	Carnegie Mellon University Graduate Research Assistant, PITTSBURGH, USA In my PhD, building scalable vision-language models through unified 2D, 3D, and 4D representations for spatial understanding and long-horizon reasoning (with Fragkiadaki).
May 2026 Present	Google Research Intern, SUNNYVALE, USA Working with Dr. Lan Wang, Prof. Ming-Hsuan Yang, and Prof. Leonidas Guibas on 4D vision-language models.
May 2025 Nov. 2025	Meta Reality Labs Research Scientist Intern, REDMOND, USA Built TrackEverything, a dense 3D point tracker that scales to long videos (1000+ frames) via persistent 3D scene tracks; first-authored paper under submission (with Harley & Zhang).
May 2024 Dec. 2024	Meta FAIR Research Scientist Intern, PITTSBURGH, USA Built UniVLG, a unified 2D–3D vision-language model with SOTA 3D referential grounding; first-author ICML 2025, plus 2 collaborative ICML papers (Locate-3D Spotlight, LIFT-GS), with Meier & Sax.
May 2022 Aug. 2022	Apple Machine Learning Research Research Intern, CUPERTINO, USA Explored unsupervised/few-shot multimodal learning by aligning unpaired modalities via Gromov–Wasserstein neighborhood matching and iterative contrastive training; with Jaitly & Bautista.

May 2020 Jul. 2021	Carnegie Mellon University Research Associate, PITTSBURGH, USA Project Page Code Paper Built a self-supervised framework where an embodied agent improves a 2D detector in unseen scenes by aggregating multi-view RGB-D detections into 3D, then re-projecting pseudo-labels; first-author BMVC 2021 (with Fragkiadaki).
Aug. 2019 May 2020	MultiCog Research Group Computer Vision Research Assistant, PILANI, India Project Page Code Paper Built an aerial object detector improving over RetinaNet by 10%; co-first-author IEEE Network 2021 (with Narang).
May 2019 Aug. 2019	Indira Gandhi Center for Atomic Research Research Assistant, CHENNAI, India Project Page Code Paper Built NText and NQuAD, a nuclear-domain language dataset with 730 QA pairs, and the Nuke-BERT pretrained model; first-author arXiv 2020 (with Meenachi).

PUBLICATIONS

* denotes equal contribution.

- [14] **Jain, A.**, Paruchuri, S., Gupta, I., Zhang, F., Schmidt, T., Engel, J., Fragkiadaki, K., Harley, A. W., 2026 “TrackEverything: Dense 3D Point Tracking in Long Videos via 3D Scene Representations”. **(In Submission)**
- [13] Lin, A.* , **Jain, A.***, Liu, Y., Fragkiadaki, K., 2026 “Qwen-3D: A Generalist 3D Vision-Language Model for Spatial Understanding”. **(ECCV 2026)**
- [12] Chittupalli, S., **Jain, A.**, Huang, D., 2026 “Pi-HOC: Pairwise 3D Human-Object Contact Estimation”. **(In Submission)**
- [11] **Jain, A.***, Swerdlow, A.* , Wang, Y., Arnaud, S., Martin, A., Sax, A., Meier, F., Fragkiadaki, K., 2025 “Unifying 2D and 3D Vision-Language Understanding”. **(ICML 2025)** [Website](#)
- [10] Sarch, G., Saha, S., Khandelwal, N., **Jain, A.**, Tarr, M. J., Kumar, A., Fragkiadaki, K., 2025 “Grounded Reinforcement Learning for Visual Reasoning”. **(NeurIPS 2025)** [Website](#)
- [9] McVay, P.* , Arnaud, S.* , Martin, A., Majumdar, A., Jatavallabhula, K. M., Thomas, P., Partsey, R., Dugas, D., Gejji, A., Sax, A., Berges, V.-P., Henaff, M., **Jain, A.**, Cao, A., Prasad, I., Kalakrishnan, M., Rabbat, M., Ballas, N., Assran, M., Maksymets, O., Rajeswaran, A., Meier, F., 2025 “Locate 3D: Real-World Object Localization via Self-Supervised Learning in 3D”. **(ICML 2025)** [Spotlight](#) [Website](#)
- [8] Cao, A., Arnaud, S., Maksymets, O., Yang, J., **Jain, A.**, Yenamandra, S., Martin, A., Berges, V.-P., McVay, P., Partsey, R., Rajeswaran, A., Meier, F., Johnson, J., Park, J. J., Sax, A., 2025 “LIFT-GS: Cross-Scene Render-Supervised Distillation for 3D Language Grounding”. **(ICML 2025)** [Website](#)
- [7] **Jain, A.**, Katara, P., Gkanatsios, N., Harley, A., Sarch, G., Aggarwal, K., Chaudhary, V., Fragkiadaki, K., 2024 “ODIN: A Single Model for 2D and 3D Segmentation”. **(CVPR 2024)** [Highlight](#) [Website](#)
- [6] Yang, B., Su, H., Gkanatsios, N., Ke, T., **Jain, A.**, Schneider, J., Fragkiadaki, K., 2024 “Diffusion-ES: Gradient-free Planning with Diffusion for Autonomous Driving and Zero-Shot Instruction Following”. **(CVPR 2024)** [Website](#)
- [5] Gkanatsios, N.* , **Jain, A.***, Zhou, X., Zhang, Y., Atkeson, C., Fragkiadaki, K., 2023 “Energy-based Models are Zero-Shot Planners for Compositional Scene Rearrangement”. **(RSS 2023)** [Website](#)
- [4] **Jain, A.***, Gkanatsios, N.* , Mediratta, I., Fragkiadaki, K., 2022 “Bottom Up Top Down Detection Transformers for Language Grounding in Images and Point Clouds”. **(ECCV 2022)** [Website](#)
- [3] Fang, Z.* , **Jain, A.***, Sarch, G.* , Harley, A., Fragkiadaki, K., 2021 “Move to See Better: Self-Improving Embodied Object Detection”. **(BMVC 2021)** [Paper](#) [Code](#)
- [2] **Jain, A.***, Ramaprasad, R.* , Narang, P., et al., 2021 “AI-Enabled Object Detection in Unmanned Aerial Vehicles for Edge Computing Applications”. **(IEEE Network 2021)** [Paper](#) [Code](#)

[1] **Jain, A.**, Meenachi, N. M., Venkatraman, B., 2020 “NukeBERT: A Pre-trained language model for Low Resource Nuclear Domain”. ([arXiv 2020](#)) [Paper](#) [Code](#)

AWARDS AND SCHOLARSHIPS

2025	Meta AI Mentorship Fellowship \$150K award for AY 2025–2026
2024	CMU Robotics Vision Fellowship \$42K award for AY 2024–2025
2023–2026	Outstanding Reviewer CVPR 2024, CVPR 2026, ICCV 2023, ICCV 2025
2021–2023	Research Scholarship Full tuition funding and stipend for my masters in robotics.
2019	Google AI Summer School 1/50 students across India selected for Google AI summer school
2019	York CVR – VISTA Vision Science Summer School 1/50 students selected worldwide
2018–2020	Institute Merit Scholarship Awarded to top 3% students for Exceptional Performance
2016–2017	KVPY Scholar A national level scholarship awarded for high research potential

ACADEMIC AND ADMINISTRATIVE EXPERIENCE

2026	Area Chair , ECCV 2026
2020–Present	Reviewer for NeurIPS, CVPR, ECCV, ICCV, ICML, ICLR, ICRA, BMVC, AAAI, TPAMI
2024	Teaching Assistant , Advanced Computer Vision, CMU
2024	Teaching Assistant , Learning for 3D Vision, CMU
2019–2020	Teaching Assistant , Artificial Intelligence, BITS Pilani
2019–2020	Teaching Assistant , Machine Learning, BITS Pilani
2019–2020	Team Leader , Microsoft Student Partner, BITS Pilani
2018–2020	Teaching Assistant , Computer Programming, BITS Pilani

OUTREACH AND VOLUNTEERING

2024–Present	Mentor - ML Collective Office Hours
2024–Present	Mentor - CMU Graduate Application Support Program
2024	Mentor - Robobuddies Mentoring Program
2023–2025	Mentor - CMU Undergrad AI Mentoring Program